CHERI ACKERMAN (NÉE FAVEL)

Chemical Engineer (MEng) | Water & Wastewater Treatment · Environmental Process R&D

Abu Dhabi, UAE · +971 58 584 5473 · cherifavel@gmail.com · linkedin.com/in/cheri-favel

PROFILE

MEng Chemical Engineer with industrial R&D experience in mine-tailings wastewater treatment, dissolved-metal removal, hydrometallurgy and electrochemical processes. Skilled in planning treatment testwork, interpreting water and analytical data, evaluating process performance and preparing technical engineering documentation. Additional practical exposure to vessel reverse-osmosis watermakers, pumps and plumbing; academic environmental engineering training and published mineral-leaching research.

CORE TECHNICAL STRENGTHS

- **Industrial water treatment: mine-tailings water, iron/zinc/manganese removal, oxidation, precipitation, electrochemical treatment, process evaluation and resource recovery.**
- **Water chemistry / separation: pH adjustment, metal speciation considerations, solid–liquid separation, sampling and analysis; understanding of UF/NF/RO principles and fouling/scaling.**
- **Process engineering: mass/energy balances, process calculations, PFDs/P&IDs, equipment-sizing support, design bases, specifications and technical reporting.**
- **Laboratory / analytical: ICP-OES, XRF, XRD, FTIR, SEM-EDS, BET, TGA; experimental design, data interpretation, kinetic modelling and test documentation.**
- **Software / equipment: advanced Excel (VBA, Power Query, PivotTables), Visio and Python; foundational Aspen Plus/HYSYS exposure; vessel RO watermaker and pump maintenance support.**

ENGINEERING EXPERIENCE

Process & R&D Engineer — Free Radical Process Design (FRPD), South Africa **Mar 2021 – Jul 2022**

- Investigated water-treatment and hydrometallurgical processes through experimental programmes, including dissolved-metal removal from mine-tailings water.
- Evaluated chemical and electrochemical treatment routes; performed mass/energy balances and process calculations; supported equipment sizing and prepared PFDs/P&IDs, design bases and reports.
- Planned laboratory testwork, assessed analytical results and process feasibility, and communicated recommendations to clients; reviewed relevant supplier quotations.

SELECTED WATER, ENVIRONMENTAL & R&D PROJECTS

- **Tailings-water treatment, Peru: Investigated removal/recovery of Fe, Zn and Mn using oxidation, chemical precipitation and electrolysis; assessed treatment performance and feasibility.**
- **Rotating-cathode zinc removal: Contributed to investigation of electrochemical zinc removal from mine-tailings water, including scale-up considerations for a mobile application.**
- **Manganese processing and recovery: Investigated selective precipitation and electrochemical beneficiation routes; supported manganese sulphate monohydrate process development.**
- **Fe/Al precipitation from leach liquor: Investigated MgO, $CaCO_3$ and limestone/eggshell as agents for removal of dissolved iron and aluminium.**
- **Cobalt electrowinning: Supported development and evaluation of a continuous electrochemical metal-recovery process.**
- **Phlogopite leaching / potassium recovery (MEng): Designed nitric-acid leaching experiments, modelled kinetics and characterised feed and residues; research published in Clays and Clay Minerals.**
- **Additional resource-recovery work: Gold-recovery reactor experiments on refinery eluate; Ni-MH battery metal-recovery feasibility; hydrotalcite-like material synthesis from leach liquor.**

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant — Cresco Consulting FZCO, UAE **Apr 2025 – Present**

- Prepare and coordinate business setup documentation, maintain records and support client communications and marketing activities.

Vessel Operations & Engineering Support — Research Vessel Rosa, USA & Bahamas **May 2023 – Apr 2025**

- Supported maintenance and troubleshooting of vessel RO watermakers, pumps and plumbing, alongside routine equipment checks and marine safety procedures; not large-scale desalination plant design.

Academic Journal Editor — CACTUS Communications (Remote) **Nov 2022 – Apr 2023**

- Edited scientific manuscripts for technical clarity, consistency and publisher formatting requirements.

Postgraduate Researcher — University of Pretoria, South Africa **Jan 2020 – Dec 2021**

- Planned and conducted mineral-leaching experiments, developed kinetic models and interpreted analytical data to evaluate process behaviour and material changes.
- Maintained traceable experimental records, guided research assistants and communicated findings through technical reports and peer-reviewed publication.

Teaching Assistant / Laboratory Intern — University of Pretoria **2017 – 2021**

- Supported laboratory instruction and practical investigations in mineral leaching, particle technology and engineering fundamentals.

EDUCATION & PUBLICATION

- MEng Chemical Engineering — University of Pretoria (2021); nitric-acid leaching kinetics and solid-residue characterisation; published in Clays and Clay Minerals.
- BEngHons Chemical Engineering, Distinction — University of Pretoria (2020); Academic Honorary Colours.
- BEng Chemical Engineering — University of Pretoria (2019); Environmental Engineering specialisation including water/effluent treatment and waste remediation.

SELECTED TRAINING & CERTIFICATIONS

- Project Management: Energy & Water Projects — PMI / LinkedIn Learning (2026); LEED Green Associate Exam Prep — LinkedIn Learning (2026).
- Occupational Safety & Health: Hazard Communication — LinkedIn Learning (2026); STCW Basic Safety Training (2023); Approved Engine Course AEC 1 (2024).

ADDITIONAL INFORMATION

South African · UAE resident visa · Available immediately · Languages: English (native), Afrikaans (proficient), Hebrew (basic)